

QUESTION 1

Discuss the model statistics and equation

Model	Descriptor	Statistics				Equation	Outliers	
		R ²	RMSE	Val R ²	Val RMSE		Training	Testing
Hajira	5	0.69	0.49	0.72	0.49	log EC50 = -3.75492 -0.02527 * SlogP_VSA1 -0.01766 * SlogP_VSA2 -0.00868 * zagreb +0.09360 * apol -0.00911 * PEOE_VSA+1	6	8
Wajiha	6	0.73	0.45	0.70	0.48	log EC50 = -4.16924 +0.04723 * apol -0.01449 * PEOE_VSA+1 -0.03695 * SlogP_VSA1 +0.00998 * SMR_VSA6 +0.01003 * Q_VSA_NEG -0.00244 * PEOE_VSA-0	5	9
Huda	4	0.71	0.47	0.68	0.49	log EC50 = -4.08755 +0.07862 * apol -0.01310 * PEOE_VSA+1 +0.00339 * SMR_VSA6 -0.03142 * SlogP_VSA1	6	10
Hina	6	0.70	0.47	0.66	0.50	log EC50 = -3.60092 +0.01060 * PEOE_VSA+0 -0.00113 * PEOE_VSA+1 +0.01251 * Q_VSA_NEG -0.02507 * Q_VSA_POL +0.01714 * SlogP_VSA0 +0.01137 * SlogP_VSA4	4	9
Hafsa	7	0.73	0.45	0.68	0.49	log EC50 = -5.70192 -0.00039*WeinerPath +0.00647*PEOE_VSA-1 +0.01163*SMR_VSA1 +0.01812 * VSA_hyd -0.01413*PEOE_VSA+1 +0.01952*SlogP_VSA4 +0.00260 *SlogP_VSA8	5	11
H.Maghaz	12	0.75	0.43	0.70	0.47	log EC50 = -6.54257 +0.02923 * SMR_VSA1 +0.00891 * Weight -0.02056 * zagreb -0.00042 * weinerPath -0.04656 * vsa_hyd +0.02601 * vdW_vol -0.04845 * TPSA +0.03033 * SMR_VSA7 +0.03933 * SMR_VSA6 +0.02419 * SMR_VSA5 +0.00254 * SMR_VSA4 +0.02609 * SMR_VSA2	6	11

Interpretation of models

Model 1:

$$\log EC50 = -3.75492 - 0.02527 * \text{SlogP_VSA1} - 0.01766 * \text{SlogP_VSA2} - 0.00868 * \text{zagreb} \\ + 0.09360 * \text{apol} - 0.00911 * \text{PEOE_VSA+1}$$

This linear regression equation predicts the log EC50 represents the concentration of a compound required to achieve half of its maximum effect, a standard measure of potency. The dependent variable is logEC50, while the independent variables are molecular descriptors representing specific structural and physicochemical properties of the compounds. The intercept, -3.75492, indicates the baseline value of logEC50 when all descriptors are zero. Each descriptor is associated with a coefficient that quantifies its contribution to the predicted value. Positive coefficients suggest an increase in the descriptor value enhances potency while negative coefficients imply the opposite effect. In this model, **SlogP_VSA1** and **SlogP_VSA2** (with coefficients -0.02527 and -0.01766, respectively) negatively influence logEC50, suggesting that higher values of these descriptors reduce potency. Similarly, **zagreb** and **PEOE_VSA+1**, with coefficients -0.00868 and -0.00911, respectively, also contribute negatively, reflecting a detrimental impact on potency when their values increase. In contrast, the descriptor **apol** (coefficient +0.09360) has a strong positive influence, indicating that greater values of this property significantly enhance the compound's predicted potency. The equation reflects how changes in specific molecular properties affect the potency of a compound, with the most significant contributions coming from **apol** (positive) and **SlogP_VSA1** (negative).

Model 2:

$$\log EC50 = -4.16924 + 0.04723 * \text{apo} - 0.01449 * \text{PEOE_VSA+1} - 0.03695 * \text{SlogP_VSA1} + 0.00998 * \text{SMR_VSA6} \\ + 0.01003 * \text{Q_VSA_NEG} - 0.00244 * \text{PEOE_VSA-0}$$

This linear regression equation predicts logEC50. Positive coefficients, such as for **apo**, **SMR_VSA6**, and **Q_VSA_NEG**, indicate that increases in these descriptors enhance potency. Conversely, negative coefficients, like those for **PEOE_VSA+1**, **SlogP_VSA1**, and **PEOE_VSA-0**, suggest that higher values reduce potency. This model reflects a balance between molecular properties, guiding optimization for improved compound efficacy.

Model 3:

$$\log EC50 = -4.08755 + 0.07862 * \text{apol} - 0.01310 * \text{PEOE_VSA+1} + 0.00339 * \text{SMR_VSA6} \\ - 0.03142 * \text{SlogP_VSA1}$$

This equation predicts logEC50 measures compound potency. Positive coefficients, such as for **apol** (+0.07862) and **SMR_VSA6** (+0.00339), indicate that increases in these descriptors enhance potency. In contrast, negative coefficients, such as for **PEOE_VSA+1** (-0.01310) and **SlogP_VSA1** (-0.03142), imply that higher values reduce potency. The equation highlights the relative influence of molecular properties, with **apol** having the strongest positive effect and **SlogP_VSA1** the most significant negative impact.

Model 4:

$$\log EC50 = -3.60092 + 0.01060 * \text{PEOE_VSA+0} - 0.00113 * \text{PEOE_VSA+1} + 0.01251 * \text{Q_VSA_NEG} - 0.02507 * \text{Q_VSA_POL} \\ + 0.01714 * \text{SlogP_VSA0} + 0.01137 * \text{SlogP_VSA4}$$

This equation predicts logEC50 represents the compound's potency. Positive coefficients, such as for **PEOE_VSA+0** (+0.01060), **Q_VSA_NEG** (+0.01251), **SlogP_VSA0** (+0.01714), and **SlogP_VSA4** (+0.01137), indicate that increases in these descriptors enhance potency. Conversely, negative coefficients for **PEOE_VSA+1** (-0.00113) and

Q_VSA_POL (-0.02507) suggest that higher values of these descriptors reduce potency. Among these, **SlogP_VSA0** has the strongest positive influence, while **Q_VSA_POL** exerts the most significant negative effect, highlighting the key factors driving compound efficacy.

Model 5:

$$\log EC50 = -5.70192 - 0.00039 * \text{Weiner Path} + 0.00647 * \text{PEOE_VSA-1} + 0.01163 * \text{SMR_VSA1} + 0.01812 * \text{VSA_hyd} - 0.01413 * \text{PEOE_VSA+1} + 0.01952 * \text{SlogP_VSA4} + 0.00260 * \text{SlogP_VSA8}$$

This linear regression model predicts the log of EC50, a measure of compound potency, based on several molecular descriptors. Each descriptor in the equation represents a specific molecular property, and its coefficient indicates the direction and magnitude of its impact on logEC50. A positive coefficient indicates that the descriptor increases the compound's potency whereas a negative coefficient indicates the opposite effect. The Weiner Path descriptor has a small negative coefficient (-0.00039), indicating a minimal reduction in potency as its value increases. The PEOE_VSA-1 descriptor has a positive coefficient (+0.00647), which means it enhances potency. Similarly, SMR_VSA1, and VSA_hyd, contribute positively to potency with coefficients of +0.01163 and +0.01812, respectively. In contrast, the PEOE_VSA+1 descriptor has a negative impact (-0.01413), reducing log(EC50) as its value increases. Lipophilicity descriptors like SlogP_VSA4 and SlogP_VSA8 show positive contributions to potency with coefficients of +0.01952 and +0.00260, respectively. Among these, SlogP_VSA4 and VSA_hyd have the strongest positive effects, while PEOE_VSA+1 exerts the most notable negative influence on logEC50 value.

Model 6:

$$\log EC50 = -6.54257 + 0.02923 * \text{SMR_VSA1} + 0.00891 * \text{Weight} - 0.02056 * \text{zagreb} - 0.00042 * \text{weinerPath} - 0.04656 * \text{vsa_hyd} + 0.02601 * \text{vdw_vol} - 0.04845 * \text{TPSA} + 0.03033 * \text{SMR_VSA7} + 0.03933 * \text{SMR_VSA6} + 0.02419 * \text{SMR_VSA5} + 0.00254 * \text{SMR_VSA4} + 0.02609 * \text{SMR_VSA2}$$

This linear regression model predicts logEC50 based on various molecular descriptors. Positive coefficients associated with SMR_VSA1, Weight, SMR_VSA2, and others, indicate that an increase in these descriptors enhances the predicted potency of the compound. While, negative coefficients zagreb, weinerPath, vsa_hyd, and TPSA, show that higher values of these descriptors reduce potency. Descriptors with larger absolute coefficients, such as TPSA (-0.04845), vsa_hyd (-0.04656), and SMR_VSA7 (+0.03033), have a stronger influence on the prediction compared to those with smaller coefficients like SMR_VSA4 (+0.00254).

QUESTION 2

Are there some outliers in your model, IF YES discuss the outlier behavior

Yes, there are outliers in the model.

Training Set:

The model has 6 outliers in the training set.

OUTLIERS CODE	XRES VALUE
GPV409	-1.7184

GPV571	-1.6211
GPV182	-1.6185
GPV220	-1.1248
GPV842	-1.0811
GPV319	1.1229

Testing Set:

The testing set also contains 8 outliers.

RESIDUAL VALUE	OUTLIERS CODE
-2.4575	3
-1.8338	5
-1.4142	1
-1.0199	10
1.5984	12
1.7496	4
1.9949	7
2.5244	13

Behavior:

1. Unique Chemistry of the Training Data:

The compounds may possess structural or chemical features not adequately represented by the descriptors. The descriptors we selected might be specific to the training data (141 compounds) only but when we test those descriptors against the test data the descriptors may not give desired results. So, the training data due to having unique chemistry both structurally and chemically may not match the test set. Both the testing and training data may belong to different congeneric classes.

2. Absence of some important or 3D/3Di descriptors:

We may have missed some important descriptors and the 2D descriptors may not be enough for better model predictions.

3. Variation in Response:

The **absence of variation** in drug responses can also lead to outliers in QSAR studies. If there is no variation in responses, then we will be unable to predict the model. So, there should be significant difference among the activity value in addition to the least and most active values. There should be at least 2 to 3 magnitude difference in activity value. When multiple compounds exhibit nearly identical activity values (e.g., identical IC₅₀, EC₅₀, or K_i values), they may appear as statistical outliers in models that assume variability. Models may misinterpret this as a lack of correlation between structure and activity, reducing their predictive power.

4. Incorrect Biological system and experiment:

To calculate the activity values there should be proper experiment and a proper biological system. So we might have ignored this when building the qsar model.

5. Balance in actives and inactives:

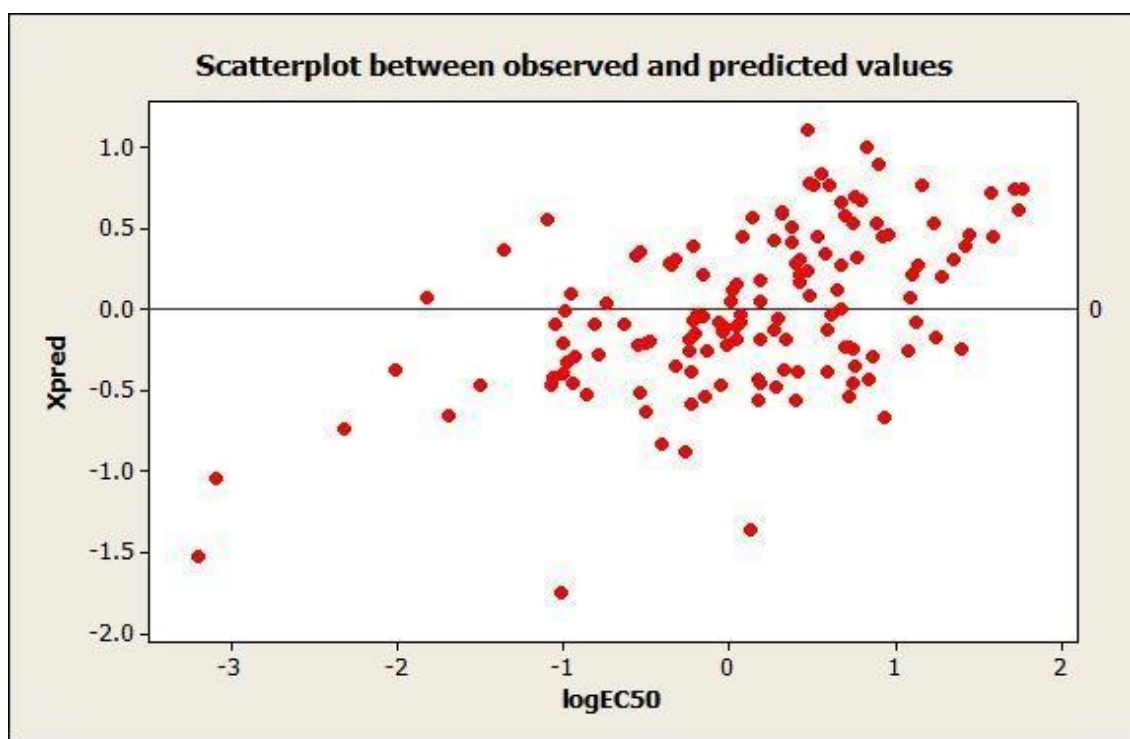
As we know there should be a good balance between actives and inactives. But in our training data there is not a good balance between them. There are a lot of actives and very few inactives. Due to this reason we may have outliers in the training data.

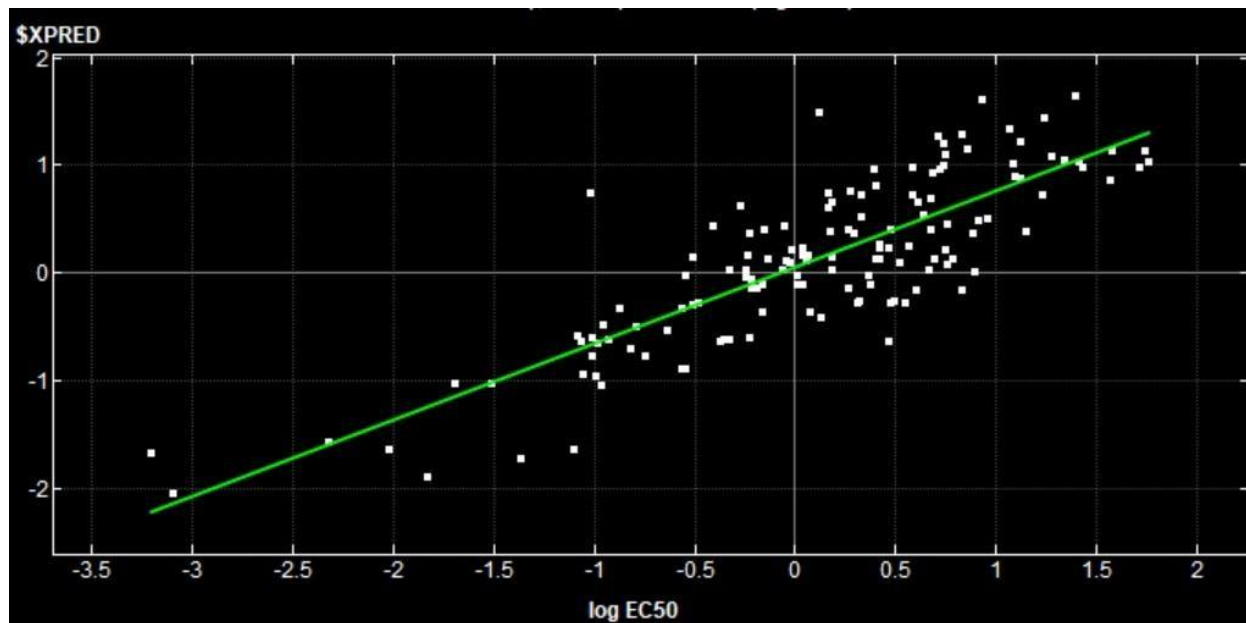
6. Applicability Domain (AD):

Outliers often fall outside the AD of the model, indicating they are not well-represented by the training set.

QUESTION 3

Provide a correlation plot between observed vs. predicted (LOO) biological activity values of the training set highlight the outliers if any.

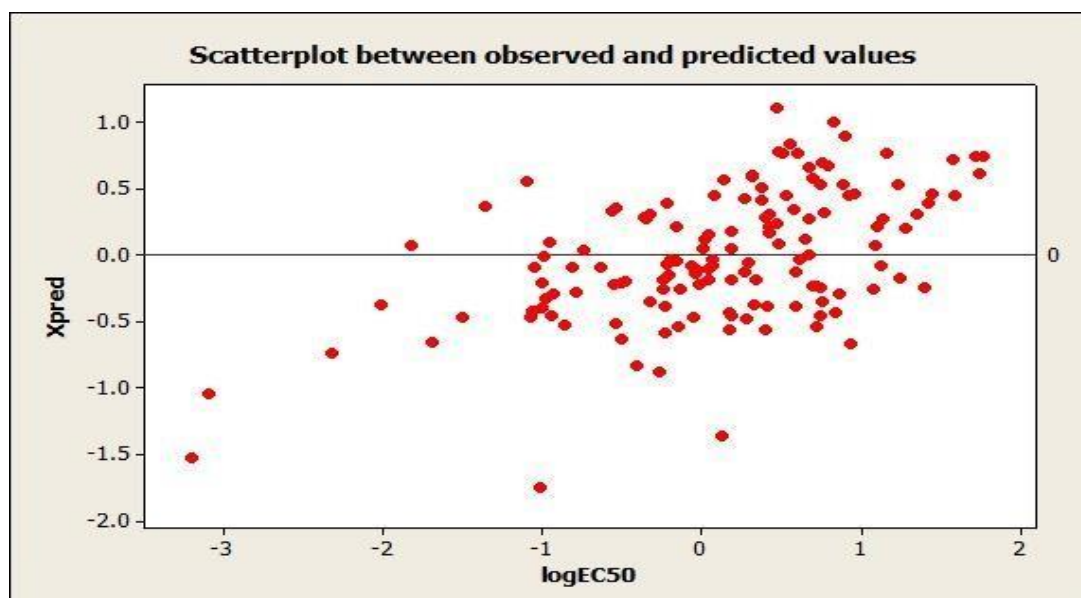


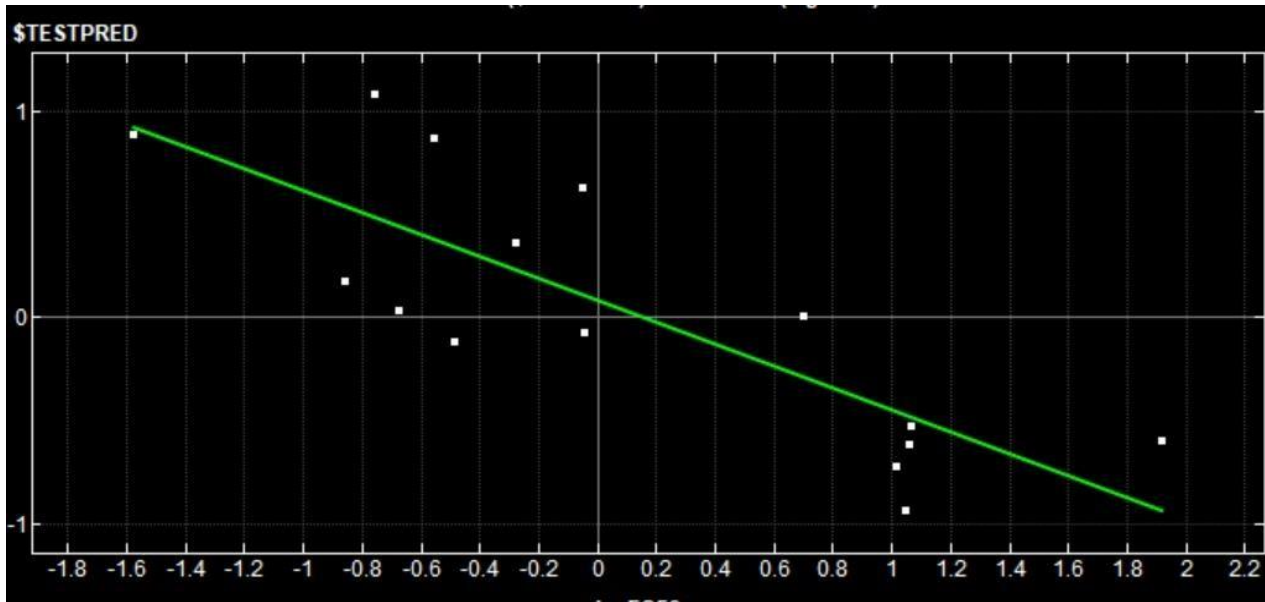


There are six outliers in the above plots with values greater than 0.1 and less than -6. The six outliers are named GPV409, GPV571, GPV182, GPV220, GPV842, GPV319 with residual values of -1.7184, -1.6211, -1.6185, -1.1248, -1.0811 and 1.1229.

QUESTION 4

Provide a correlation plot of observed vs. predicted biological activities of the test





There are outliers in plot with values above 0.1 and below -1. The outlier's residual has values of -2.4575, -1.8338, -1.4142, -1.4142, -1.0199, 1.5984, 1.7496, 1.9949 and 2.5244.

QUESTION 5 & 6

Provide the SVL model report and Provide the fit file.

The SVL model reports and fit files are provided in a folder.